

# FPGA-Based Traffic Light Controller with Pedestrian and Emergency Vehicle Priority

## Abstract

This project presents the design and verification of an FPGA-Based Traffic Light Controller implemented using SystemVerilog. The controller uses a Finite State Machine (FSM) to manage North-South and East-West traffic flow while supporting pedestrian crossing requests and emergency vehicle priority. The design was verified through simulation using Icarus Verilog and GTKWave and is suitable for FPGA implementation.

## Objective

To design, simulate, verify, and implement a robust traffic light controller using FPGA technology and FSM-based digital design principles.

## System Architecture

Inputs: Clock, Reset, Pedestrian Request, Emergency Vehicle Request. Processing Blocks: FSM Controller, Timer Counter, Priority Logic. Outputs: NS\_R, NS\_Y, NS\_G, EW\_R, EW\_Y, EW\_G, PED\_WALK.

## FSM Design

States: S0 - NS Green S1 - NS Yellow S2 - EW Green S3 - EW Yellow S4 - All Red S5 - Pedestrian Walk Normal Flow: S0 -> S1 -> S2 -> S3 -> S0 Pedestrian Flow: S0/S2 -> Yellow -> All Red -> Pedestrian Walk -> S0 Emergency Flow: Any State -> S0 (NS Green Priority)

## State Table

000 = NS Green 001 = NS Yellow 010 = EW Green 011 = EW Yellow 100 = All Red 101 = Pedestrian Walk

## RTL Design

The RTL was implemented using SystemVerilog. The design uses `always_ff` for sequential state transitions and `always_comb` for output generation. A timer controls state duration while request signals drive event-based transitions.

## Testbench Design

A dedicated testbench generates clock and reset signals, injects pedestrian requests, triggers emergency vehicle conditions, and records simulation activity into a VCD waveform file.

## Waveform Analysis

Simulation confirms correct state sequencing, timer operation, pedestrian handling, emergency priority override, and safe traffic signal behavior. The waveform demonstrates successful transitions through all FSM states.

## FPGA Implementation

Implementation Steps: 1. Create Vivado Project 2. Add RTL Source Files 3. Add Constraints File 4. Run Synthesis 5. Run Implementation 6. Generate Bitstream 7. Program FPGA Traffic signals can

be mapped to FPGA LEDs for hardware demonstration.

## **Results**

The controller successfully performs normal traffic control, pedestrian crossing management, and emergency vehicle prioritization. Simulation results verify correct operation of all FSM states and outputs.

## **Applications**

Smart Traffic Systems, Smart Cities, Industrial Automation, Embedded Controllers, Transportation Infrastructure, and FPGA-Based Control Systems.

## **Conclusion**

The FPGA-Based Traffic Light Controller demonstrates FSM design, RTL development, verification methodology, and FPGA implementation concepts. The enhanced design with pedestrian and emergency support provides industry-level complexity and strong portfolio value for VLSI and FPGA roles.